Patterns, proofs & practice

A mixed problem set for algebra, geometry, and early calculus

Name _____ Date _____

1. Rational equations

Solve the equation. State the excluded value and check your solution.

$$\frac{3x - 2}{x + 1} = 2 + \frac{1}{x + 1} \qquad x \neq -1$$

2. Quadratic structure

Factor the function, identify its zeros, and write it in completed-square form.

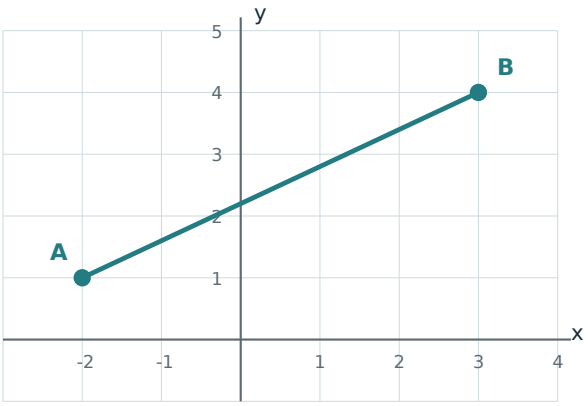
$$f(x) = x^2 - 6x + 5$$

Zeros: _____ Vertex: _____

3. A line through two points

The diagram shows A(−2, 1) and B(3, 4).

(a) Find the slope of line AB.
(b) Write an equation of line AB.
(c) Find its y-intercept.



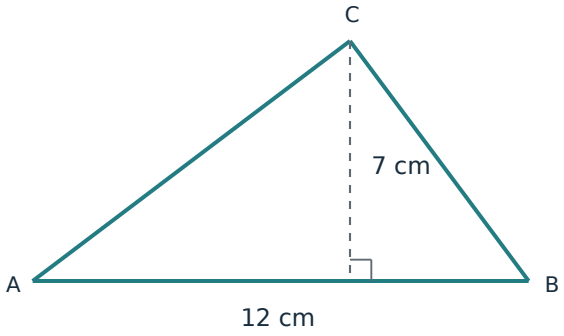
Reason across representations

Work down each column. Label every unit and give exact answers where possible.

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4. Area from a diagram

Find the area of triangle ABC. Explain which length is the perpendicular height.



5. Simplify with care

Simplify the expression and list all values excluded by the original denominator.

$$\frac{x^2 - 9}{x^2 + 3x}$$

6. Accumulated change

Evaluate the definite integral. Show an antiderivative and your substitution.

$$\int_0^2 (3x^2 + 1)\,dx$$

7. A small data story

Six students recorded the number of books they read in a month.

2 3 3 4 6 6

(a) Find the mean and median.

(b) A seventh student read 11 books. Which measure changes more? Explain.

Connect rates and totals

Read the model carefully. Question 9 continues on the scanned fourth page.

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8. A curve and its derivative

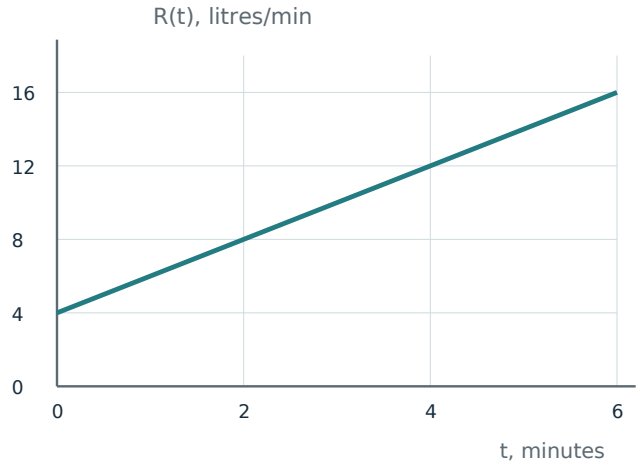
For $g(x) = x^3 - 3x$, find $g'(x)$ and all stationary points. Then evaluate:

$$\int_1^3 x^2 \, dx$$

Give the exact value as a fraction.

9. A tank, a rate, and a decision

A tank initially contains 12 litres of water. Water enters at the rate $R(t) = 4 + 2t$ litres per minute, where t is measured in minutes and $0 \le t \le 6$. No water leaves the tank.



(a) Use the graph or an integral to find the amount of water added during the first 3 minutes.

Explain why your answer differs from the total water in the tank at $t = 3$.

Continue Question 9 on page 4. Keep the same tank model.

Follow the flow

A scanned supplement: continue the model, then solve the geometry question.

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9. Tank model (continued)

Use R(t) = 4 + 2t litres per minute and the initial volume V(0) = 12 litres from page 3.

(b) Write a formula for V(t), the total volume in the tank after t minutes.

(c) Find the first time the tank contains 60 litres. Check that your time lies within $0 \le t \le 6$.

10. Geometry from a scan

A circle has radius 6 cm. A sector has central angle 60°.

(a) Find the length of the minor arc.
(b) Find the area of the sector.

Give both answers in terms of π.

